

1 Answer:

The processes run as follows :(each cell represents the running time of 5s)

Process A:

cpu	cpu	a	wait	wait	wait	cpu	b	b	wait	cpu
cpu										

Process B:

a	a	wait	wait	cpu	cpu	b	wait	wait	cpu	wait
b	b									

Process C:

b	wait	cpu	cpu	a	wait	wait	cpu	cpu	b	b

CPU utilization: $60/65=92.3\%$.

2 Answer:

1. The Need matrix is:

$$\begin{array}{ccc} 0 & 0 & 4 \\ 1 & 7 & 5 \\ 2 & 3 & 5 \\ 0 & 6 & 4 \\ 0 & 6 & 5 \end{array} \quad \begin{array}{c} - \\ \\ \\ \\ \end{array} \quad \begin{array}{ccc} 0 & 0 & 3 \\ 1 & 0 & 0 \\ 1 & 3 & 5 \\ 0 & 0 & 2 \\ 0 & 0 & 1 \end{array} \quad = \quad \begin{array}{ccc} 0 & 0 & 1 \\ 0 & 7 & 5 \\ 1 & 0 & 0 \\ 0 & 6 & 2 \\ 0 & 6 & 4 \end{array}$$

First, the Available is (2,4,0), which can meet the needs of P2. After P2 finished, the Available is (3,7,5), which can meet the needs of any process, so the system is in a safe state.

2. No. If the Available is (0,5,3), it can meet the needs of P0; When P0 finished, the Available is (0,5,6). At this time, the Available can't meet the needs of any process, and the system will have a deadlock.